

Recursion : Sum of n numbers:

```
Sum(n)
{
  int sum=0;
  if
  return
```

```
#include <stdio.h>
int main() {
  int n;
  Sum(n);
  return 0;
}
```

```
int Sum(int a)
{
  int sum=0;
  if(a==1) {
    return 1;
  }
  return sum(a-1)+a;
}
```

- ⊙ find out the Applications of recursion
- ⊙ Is recursion still used in the programming era why
- ⊙ What are the programming languages that support recursive calls.

1) Recursion is a programming technique in which a function calls itself.

1) Tree Traversal: Recursion is commonly used to traverse tree data structures such as binary trees.

ex: InOrder traversal

preOrder traversal

postOrder traversal

2) Searching and Sorting Algorithms: Many efficient algorithms use recursion.

ex:- Binary search
Merge sort